



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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**INTELLIGENT VACCINE COLD STORAGE MONITORING SYSTEM
INDUSTRY CONTEXT**

ARCHITECTURE DESIGN ANALYSIS REPORT

Submitted in the partial fulfilment for the Course of

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to the award of the degree of

BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY

Submitted by

SAMEER AHMED G (192421154)

Under the Supervision of

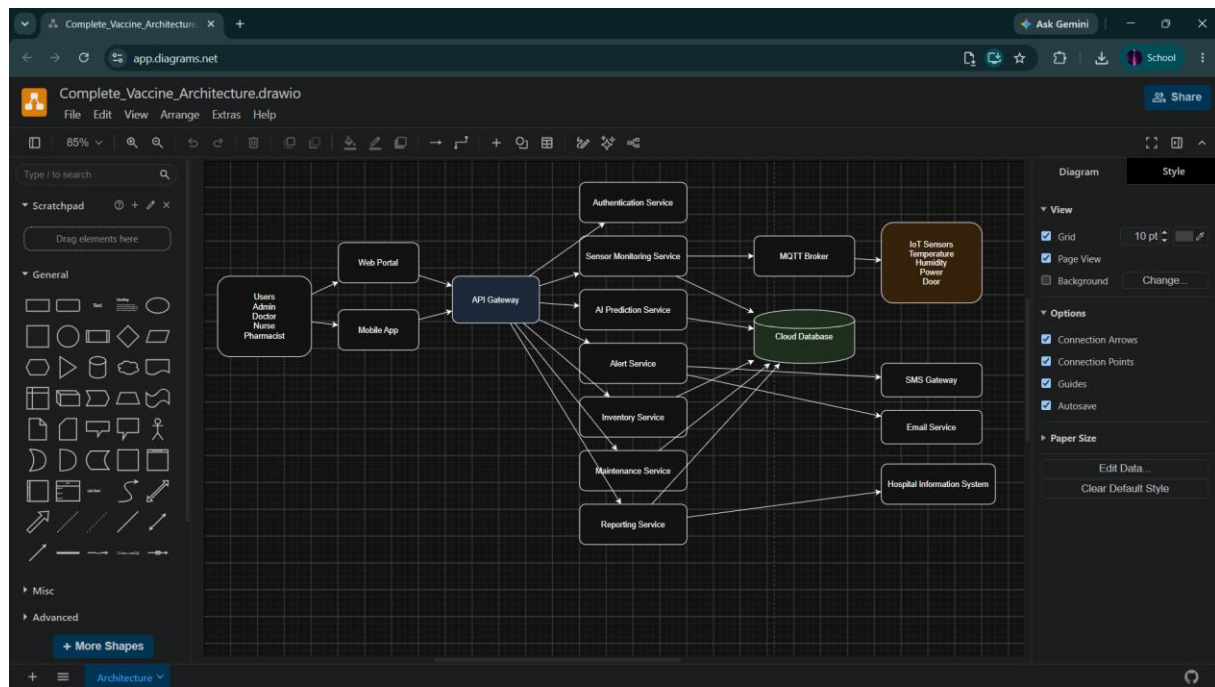
DR ANITA DAVAMANI

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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1. SYSTEM REQUIREMENT ANALYSIS

1.1 System Objectives

The Intelligent Vaccine Cold Storage Monitoring System is developed to ensure safe storage of vaccines by continuously monitoring environmental conditions using IoT sensors. The system predicts refrigeration failures, generates instant alerts, stores historical records, and provides automated compliance reports. It improves vaccine safety while reducing operational costs and minimizing vaccine wastage.

Objectives

- Monitor temperature continuously.
- Monitor humidity.
- Detect abnormal storage conditions.
- Predict refrigeration failures.
- Generate alerts instantly.
- Maintain audit logs.
- Produce compliance reports.
- Improve vaccine safety.
- Support multiple healthcare facilities.

1.2 Functional Requirements

The proposed software shall:

- Authenticate users.
- Monitor temperature and humidity.
- Collect IoT sensor data.
- Display live dashboard.
- Predict equipment failures.
- Send SMS and Email alerts.
- Generate reports.
- Maintain inventory records.
- Schedule maintenance.
- Store historical data.
- Support multiple hospitals.

1.3 Non-Functional Requirements

The system should satisfy the following quality requirements.

Performance

- Response time below 2 seconds.
- Real-time sensor updates.

Security

- HTTPS communication.
- Role-Based Access Control.
- Data encryption.

Reliability

- Automatic backup.
- Continuous monitoring.
- Fault tolerance.

Scalability

- Support multiple hospitals.
- Support thousands of sensors.

Maintainability

- Modular architecture.
- Reusable services.
- Easy software upgrades.

Availability

- 99.9% uptime.
- Cloud deployment.

1.4 Quality Attributes

The architecture is designed to satisfy important software quality attributes including:

- Scalability
- Security
- Reliability
- Performance
- Maintainability
- Availability
- Flexibility

These attributes influenced the architectural decisions during system design.

2. SELECTION OF SOFTWARE ARCHITECTURE

Selected Architecture

Hybrid Microservices Architecture

The proposed system adopts a **Hybrid Microservices Architecture** that combines Layered Architecture, Microservices, and Event-Driven communication.

The architecture contains independent services such as Authentication, Sensor Monitoring, AI Prediction, Inventory Management, Notification Service, Maintenance Service, and Reporting Service. Each service performs a dedicated responsibility while communicating through REST APIs and MQTT messaging.

Why Hybrid Microservices?

This architecture was selected because the system:

- Uses IoT devices.
- Requires cloud deployment.
- Needs independent scaling.

- Supports multiple hospitals.
- Requires high availability.
- Integrates Artificial Intelligence.
- Generates real-time alerts.
- Supports future expansion.

Compared to Layered Architecture alone, Microservices provide better scalability and maintainability for large healthcare systems.

3. SOFTWARE ARCHITECTURE DIAGRAM

(Insert the Draw.io Architecture Diagram here.)

Figure 3.1: Software Architecture of Intelligent Vaccine Cold Storage Monitoring System

The architecture consists of:

- Client Layer
- API Gateway
- Authentication Service
- Sensor Monitoring Service
- AI Prediction Service
- Alert Service
- Inventory Service
- Maintenance Service
- Reporting Service
- Cloud Database
- MQTT Broker
- IoT Sensors
- SMS Gateway
- Email Service
- Hospital Information System

4. COMPONENTS AND THEIR INTERACTIONS

Client Layer

The Client Layer consists of the Web Portal and Mobile Application. Healthcare administrators, doctors, nurses, and pharmacists access the system through these interfaces to monitor storage conditions, receive alerts, and generate reports.

API Gateway

The API Gateway acts as the single entry point for all client requests. It authenticates requests, routes them to the appropriate microservices, and improves security by preventing direct access to internal services.

Authentication Service

This service manages user authentication, authorization, and role-based access control. Only authorized users can access system resources based on their assigned roles.

Sensor Monitoring Service

The Sensor Monitoring Service receives real-time data from IoT sensors through the MQTT Broker. It validates temperature, humidity, power supply, and equipment status before storing the information in the cloud database.

AI Prediction Service

The AI Prediction Service analyzes historical and real-time sensor data to identify abnormal operating patterns. Machine Learning algorithms predict refrigeration failures and calculate equipment health scores.

Alert Service

Whenever abnormal environmental conditions are detected, the Alert Service immediately generates SMS, Email, and Mobile Push Notifications to healthcare personnel.

Inventory Service

This service maintains vaccine inventory, tracks expiry dates, and records stock movement.

Maintenance Service

The Maintenance Service schedules preventive maintenance, stores repair history, and notifies engineers about upcoming maintenance activities.

Reporting Service

The Reporting Service generates compliance reports, audit logs, maintenance reports, and historical performance summaries required by regulatory authorities.

Cloud Database

The Cloud Database securely stores:

- User information
- Sensor readings
- Equipment status
- Inventory data
- Reports
- Maintenance history
- Audit logs

MQTT Broker

The MQTT Broker manages communication between IoT sensors and cloud services. It enables reliable and lightweight message transmission.

IoT Sensors

The sensors continuously monitor:

- Temperature
- Humidity
- Power Supply
- Door Status
- Refrigerator Health

System Workflow

1. IoT Sensors collect environmental data.
2. MQTT Broker transmits data.
3. Sensor Monitoring Service processes readings.
4. Data is stored in the Cloud Database.
5. AI Prediction analyzes sensor history.
6. Alert Service sends notifications.
7. Dashboard displays real-time information.
8. Reports are generated for administrators.

5. QUALITY ATTRIBUTE ANALYSIS

Scalability

Each microservice operates independently and can be scaled horizontally. Additional hospitals and IoT devices can be added without affecting existing services.

Security

Security mechanisms include:

- HTTPS
- JWT Authentication
- TLS Encryption
- Role-Based Access Control
- Audit Logging

These protect healthcare information from unauthorized access.

Performance

The architecture uses independent services and MQTT communication, enabling low-latency sensor monitoring and fast dashboard updates.

Reliability

Reliability is achieved through:

- Cloud deployment
- Database backup
- Fault tolerance
- Service isolation
- Disaster recovery

Maintainability

Since every module is independent, developers can update one service without affecting the remaining components. Modular architecture simplifies testing and debugging.

Availability

Cloud deployment ensures high availability through server redundancy, automatic backups, and failover mechanisms, enabling continuous monitoring even during hardware failures.

6. ARCHITECTURAL JUSTIFICATION

The Hybrid Microservices Architecture was selected because it satisfies all functional and non-functional requirements of the Intelligent Vaccine Cold Storage Monitoring System. The architecture supports modular development, real-time communication, AI integration, and cloud scalability.

Advantages

- Independent deployment of services.
- Better scalability.
- Improved fault isolation.
- Easier maintenance.
- Faster software updates.
- Supports AI and IoT integration.
- High system availability.
- Better security.

Trade-offs

- Higher implementation complexity.
- Increased deployment cost.
- Service communication overhead.
- More infrastructure management.

Despite these trade-offs, the long-term benefits significantly outweigh the disadvantages.

Future Enhancements

The architecture can be expanded by integrating:

- Blockchain for secure audit logs.
- Edge AI for local prediction.
- Digital Twin technology.
- Computer Vision monitoring.
- ERP integration.
- Hospital Information System integration.
- Smart Energy Management.
- Advanced Deep Learning models.

The modular architecture allows these enhancements without major redesign.

7. CONCLUSION

The Hybrid Microservices Architecture provides a scalable, secure, reliable, and maintainable solution for the Intelligent Vaccine Cold Storage Monitoring System. By combining IoT, Artificial Intelligence, Cloud Computing, MQTT communication, and independent microservices, the architecture supports real-time monitoring, predictive maintenance, automated reporting, and secure healthcare operations. This architecture is well suited for large-scale deployment across hospitals, clinics, pharmacies, and vaccination centers while supporting future technological advancements.

8. REFERENCES (IEEE FORMAT)

- [1] I. Sommerville, *Software Engineering*, 10th ed., Pearson, 2016.
- [2] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed., McGraw-Hill, 2020.
- [3] M. Richards and N. Ford, *Fundamentals of Software Architecture*. O'Reilly Media, 2020.
- [4] A. Bahga and V. Madisetti, *Internet of Things: A Hands-On Approach*. Universities Press, 2015.
- [5] M. Kleppmann, *Designing Data-Intensive Applications*. O'Reilly Media, 2017.
- [6] World Health Organization, *Temperature Sensitivity of Vaccines*, Geneva, Switzerland, 2006.
- [7] Centers for Disease Control and Prevention, *Vaccine Storage and Handling Toolkit*, 2023.
- [8] ISO/IEC 27001:2022, *Information Security Management Systems*, ISO, Geneva, Switzerland.
- [9] ISO 13485:2016, *Medical Devices – Quality Management Systems*, ISO, Geneva, Switzerland.
- [10] IEEE Computer Society, *IEEE Software Engineering Standards Collection*, IEEE, 2019.